Thesis - I Abstract

Title: Edge AI-Based Smart Campus Information Robot using Small Language Models (SLMs)

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Abstract

Modern university campuses are becoming increasingly complex, making it difficult for new students, visitors, and staff to access information and navigate facilities efficiently. Traditional information desks are limited by availability and require continuous human intervention. This project proposes the development of an Edge AI-Powered Smart Campus Information Robot that utilizes Small Language Models (SLMs) to provide real-time, intelligent, and privacy-preserving assistance within a campus environment.

The proposed robot will function as an interactive campus guide capable of answering queries related to classrooms, laboratories, faculty offices, academic departments, events, timetables, and other campus services through natural language conversations. We can use this robot for other applications as well. Unlike cloud-dependent AI assistants, the system will perform language inference locally using lightweight SLMs, enabling faster response times, reduced latency, lower operational costs, enhanced privacy, and reliable offline functionality.

The system integrates edge computing, Natural Language Processing (NLP), speech recognition, text-to-speech, computer vision, and robotics. A Retrieval-Augmented Generation (RAG) framework will connect the language model to a structured campus knowledge base, ensuring responses remain accurate and up-to-date. The robot will also employ computer vision and sensor-based navigation techniques to autonomously move within the campus while avoiding obstacles. Additional features such as multilingual support, QR-code-based navigation, touch-screen interaction, and personalized guidance can further enhance user experience.

This project aims to demonstrate the feasibility of deploying intelligent conversational robots using lightweight language models on resource-constrained edge devices. The proposed solution contributes to the vision of smart campuses by providing accessible, scalable, secure, and efficient information services while showcasing the practical integration of artificial intelligence, embedded systems, and autonomous robotics.

Keywords

Edge AI, Small Language Models (SLMs), Smart Campus, Robotics, Retrieval-Augmented Generation (RAG), Computer Vision, Edge Computing, Autonomous Navigation.